LIVE

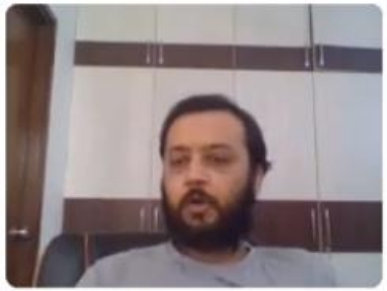
Forced Oscillations / Driven Oscillator

⇒ Forced oscillations happen when an external force compensates for the loss in the energy of a damped oscillator

⇒ A special case exists when the force itself becomes harmonic

$$F = F_0 \cos(\omega t)$$

ω = Angular frequency of driving force



Pratham omkar Pat... : good morning sir

A Hari kumar : yes

Mohit Singh : Yes

Manoj : Yes

Abhiharsh Chauhan : ya

Jatin Gupta : Good morning sir, sorry for being late....

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Chat | Doubts | More

LIVE

$$F_{net} = -kx - bv + F_0 \cos(\omega t)$$

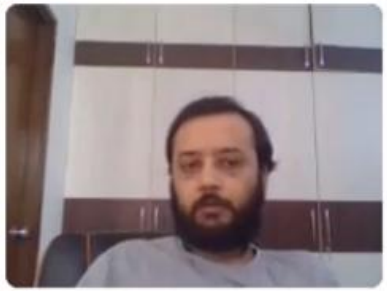
$$ma = -kx - bv + F_0 \cos(\omega t)$$

$$m\frac{d^2x}{dt^2} + b\frac{dx}{dt} + kx = F_0 \cos(\omega t)$$

$$x = A \sin(\omega t + \phi)$$

$A =$

ω = frequency of oscillation which is same as that of force F



Pratham omkar Pat... : good morning sir

A Hari kumar : yes

Mohit Singh : Yes

Manoj : Yes

Abhiharsh Chauhan : ya

Jatin Gupta : Good morning sir, sorry for being late....

Type a message

Chat Doubts More

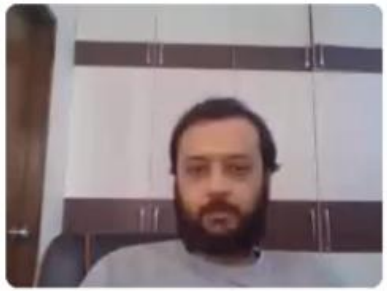
LIVE



$V_0 \cos \omega t$

$$V_0 \cos \omega t - L\frac{di}{dt} - \frac{q}{C} - iR = 0$$

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{C}q = V_0 \cos \omega t$$



AR Arnab Goyal
capacitor

RI Rishav Raj
LC osc

SA Sanyam kumar Jain
inductor

PR Pratham omkar Pat...
motional emf

AB Abhiharsh Chauhan
electromagnetic damping

AR Arnab Goyal
sir jab rod gir rahi hoti hain

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Chat Doubts More

LIVE

$$m\frac{d^2x}{dt^2} + b\frac{dx}{dt} + kx = F_0\cos(\omega t)$$

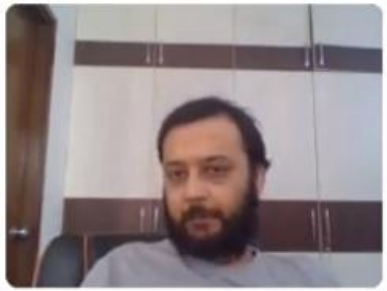
$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{C}q = V_0\cos\omega t$$

$$q = Q_0\sin(\omega t + \phi)$$

$$i = Q_0\omega\cos(\omega t + \phi)$$

$$i = \frac{V_0}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}\cos(\omega t + \phi)$$

$$\tan\phi = \frac{X_C - X_L}{R}$$



RI Rishav Raj: oo..then we can compare

A A Hari kumar: y

PR Prathik: y

RI Rishav Raj: .

MA Manoj: yes

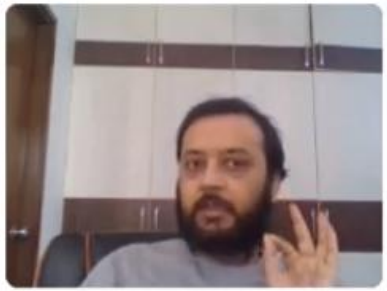
AB Abhiharsh Chauhan: y

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Chat Doubts More

$$Q_0 = \frac{V_0}{\omega \sqrt{R^2 + \left(\frac{1}{\omega C} - \omega L\right)^2}}$$

$$q = Q_0 \sin(\omega t + \phi)$$



oo..then we can compare

A Hari kumar: y

Prathik: y

Rishav Raj: .

Manoj: yes

Abhiharsh Chauhan: y

A Hari kumar: y

Type a message

Chat Doubts More

LIVE

$$Q_0 = \frac{V_0}{\omega\sqrt{R^2 + \left(\frac{1}{\omega C} - \omega L\right)^2}}$$

$$M \equiv L$$

$$b \equiv R$$

$$k \equiv \frac{1}{C}$$

$$F_0 \equiv V_0$$

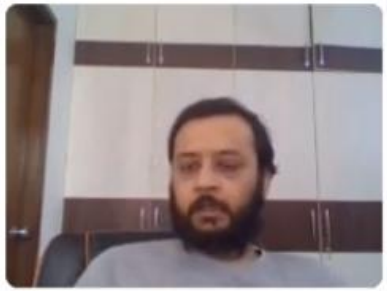
$$q = Q_0 \sin(\omega t + \phi)$$

$$x = A \sin(\omega t + \phi)$$

$$k = m\omega_0^2$$

$$A = \frac{1}{\omega}\frac{F_0}{\sqrt{b^2 + \left(\frac{k}{\omega} - \omega m\right)^2}} = \frac{F_0}{\sqrt{b^2\omega^2 + (m\omega_0^2 - m\omega^2)^2}}$$

$$A = \frac{F_0}{\sqrt{b^2\omega^2 + m^2(\omega^2 - \omega_0^2)^2}}$$



yes

AB Abhiharsh Chauhan
y

RI Rishav Raj
.

AB Abhiharsh Chauhan
ya

MA Manoj
yes

PR Pratham omkar Pat...
one more sq sir
in w^2-w0^2 ryt

RI Rishav Raj
haa

Type a message

Chat Doubts More

Condition for maximum displacement amplitude

$$A = \frac{F_0}{\sqrt{b^2\omega^2 + m^2(\omega^2 - \omega_0^2)^2}}$$

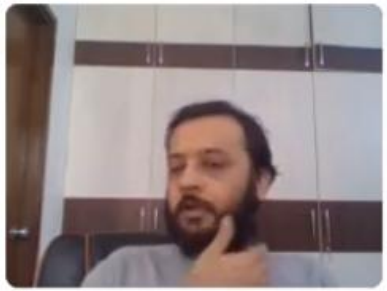
$$D^2 = b^2\omega^2 + m^2(\omega^2 - \omega_0^2)^2$$

$$\frac{d(D^2)}{d\omega} = 2\omega b^2 + m^2 2(\omega^2 - \omega_0^2)2\omega = 0$$

$$b^2 = 2m^2(\omega_0^2 - \omega^2)$$

$$\omega = \sqrt{\omega_0^2 - \frac{b^2}{2m^2}}$$

learn



Rishav Raj
haa waise hi bola maine

AB Abhiharsh Chauhan
y

MA Madhav
this is same as underdamped osccillation frreq

RI Rishav Raj
omega=0 bhi critical pt hai sir to wahan min ho rha hoga?

MA Madhav
Oh ya sry sir

Type a message

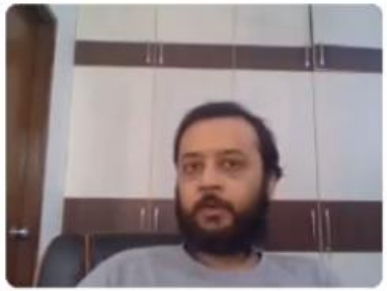
Chat Doubts More

LIVE

For small damping $b \approx 0$

$\omega \approx \omega_0$

For significant damping $\omega < \omega_0$



haa waise hi bola maine

AB Abhiharsh Chauhan
y

MA Madhav
this is same as underdamped osccillation frreq

RI Rishav Raj
omega=0 bhi critical pt hai sir to wahan min ho rha hoga?

MA Madhav
Oh ya sry sir

Type a message

Chat Doubts More

Condition for max. energy of the system

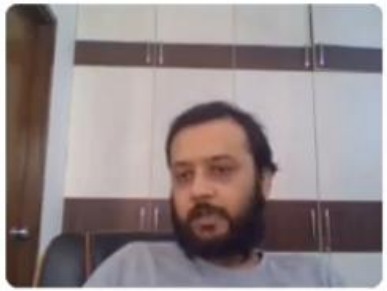
LIVE

Energy of the system

$$E = \frac{1}{2} m \omega^2 A^2$$

$$= \frac{1}{2} m \omega^2 \left(\frac{F_0^2}{b^2 \omega^2 + m^2 (\omega_0^2 - \omega^2)^2} \right)$$

$$E = \frac{1}{2} m \left[\frac{F_0^2}{b^2 + m^2 \left(\frac{\omega_0^2}{\omega} - \omega \right)^2} \right]$$



Madhav
Oh ya sry sir

AB Abhiharsh Chauhan
no

RI Rishav Raj
ni

MA Manoj
no

JA Jatin Gupta
Yes sir

AR Arnab Goyal
w=w0 for energy

JA Jashan Grover Insan
yes

Type a message

Chat Doubts More

LIVE

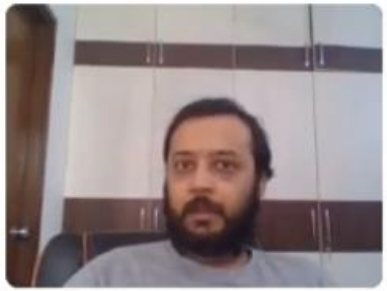
Energy of the system

$$E = \frac{1}{2} m\omega^2 A^2$$

$$= \frac{1}{2} m \omega^2 \left(\frac{F_0^2}{b^2\omega^2 + m^2(\omega_0^2 - \omega^2)^2} \right)$$

$$E = \frac{1}{2} m \left[\frac{F_0^2}{b^2 + m^2\left(\frac{\omega_0^2}{\omega} - \omega\right)^2} \right]$$

for Emax $\Rightarrow \frac{\omega_0^2}{\omega} = \omega \Rightarrow \omega_0 = \omega$



MA Manoj: no

JA Jatin Gupta: Yes sir

AR Arnab Goyal: w=w0 for energy

JA Jashan Grover insan: yes

RI Rishav Raj: sir k= mw0 kyu ni hua?

RI Rishav Raj: oo...ok sir

Type a message

Chat Doubts More

LIVE

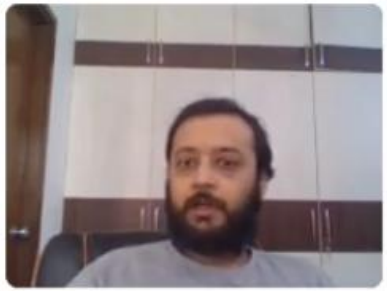
Energy of the system

$$E = \frac{1}{2} m\omega^2 A^2$$

$$= \frac{1}{2} m \omega^2 \left(\frac{F_0^2}{b^2\omega^2 + m^2(\omega_0^2 - \omega^2)^2} \right)$$

$$E = \frac{1}{2} m \left[\frac{F_0^2}{b^2 + m^2\left(\frac{\omega_0^2}{\omega} - \omega\right)^2} \right]$$

for E_{max} $\Rightarrow$ $\frac{\omega_0^2}{\omega} = \omega \Rightarrow \omega_0 = \omega$



MA Manoj: no

JA Jatin Gupta: Yes sir

AR Arnab Goyal: w=w0 for energy

JA Jashan Grover insan: yes

RI Rishav Raj: sir k= mw0 kyu ni hua?

RI Rishav Raj: oo...ok sir

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Chat Doubts More

LIVE

Energy of the system

$$E = \frac{1}{2} m (v_{max})^2$$

$$E = \frac{1}{2} m \omega^2 A^2$$

$$= \frac{1}{2} m \omega^2 \left(\frac{F_0^2}{b^2 \omega^2 + m^2 (\omega_0^2 - \omega^2)^2} \right)$$

$$E = \frac{1}{2} m \left[\frac{F_0^2}{b^2 + m^2 \left(\frac{\omega_0^2}{\omega} - \omega \right)^2} \right]$$

for E_{max} $\Rightarrow$ $\frac{\omega_0^2}{\omega} = \omega$ $\Rightarrow$ $\omega_0 = \omega$



Manoj: no

Jatin Gupta: Yes sir

Arnab Goyal: w=w0 for energy

Jashan Grover insan: yes

Rishav Raj: sir k= mw0 kyu ni hua?

Rishav Raj: oo...ok sir

Type a message

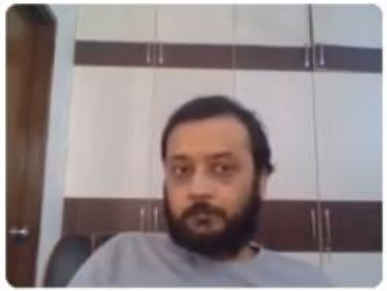
Chat Doubts More

LIVE

$\omega_0 = 8$

$\omega = 5$ $\quad E = 10\,J$ $\quad V = A\omega = 10\omega$

$\omega = 8$ $\quad E = 20\,J$ $\quad V = A\omega = 20\omega$



Rishav Raj
oo...ok sir

PR Pratham omkar Pat...
y

Jatin Gupta

PR Prathik
yes

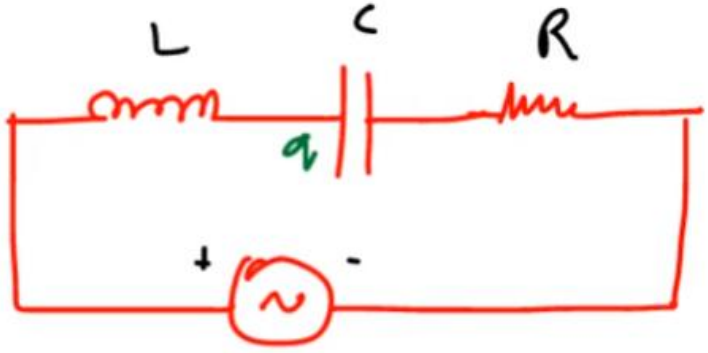
MA Manoj
yes

JA Jashan Grover insan
yes

AB Abhiharsh Chauhan
y

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Chat Doubts More

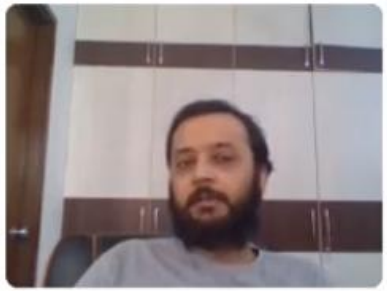


$x \equiv q$

$V \equiv i$

$$V_0 \cos \omega t - L\frac{di}{dt} - \frac{q}{C} - iR = 0$$

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{C}q = V_0 \cos \omega t$$



Prathik
yes

Manoj
yes

Jashan Grover Insan
yes

Abhiharsh Chauhan
y

Rishav Raj
.

Pranesh K.r
yes

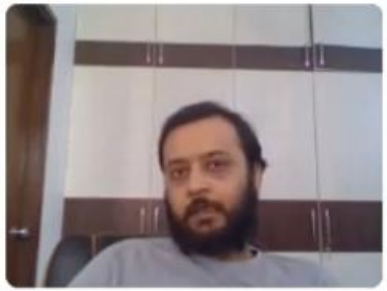
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Chat Doubts More

LIVE

Resnick: When the forced oscillation happen at natural frequency the velocity amplitude & energy becomes max. $\omega = \omega_0$

NCERT: When the angular frequency of oscillator approaches the natural frequency then displacement Amplitude becomes max; a condition called as Resonance



Abhiharan Chauhan
y

RI Rishav Raj
.

JA Jashan Grover Insan
chmkaaa

MA Manoj
yes

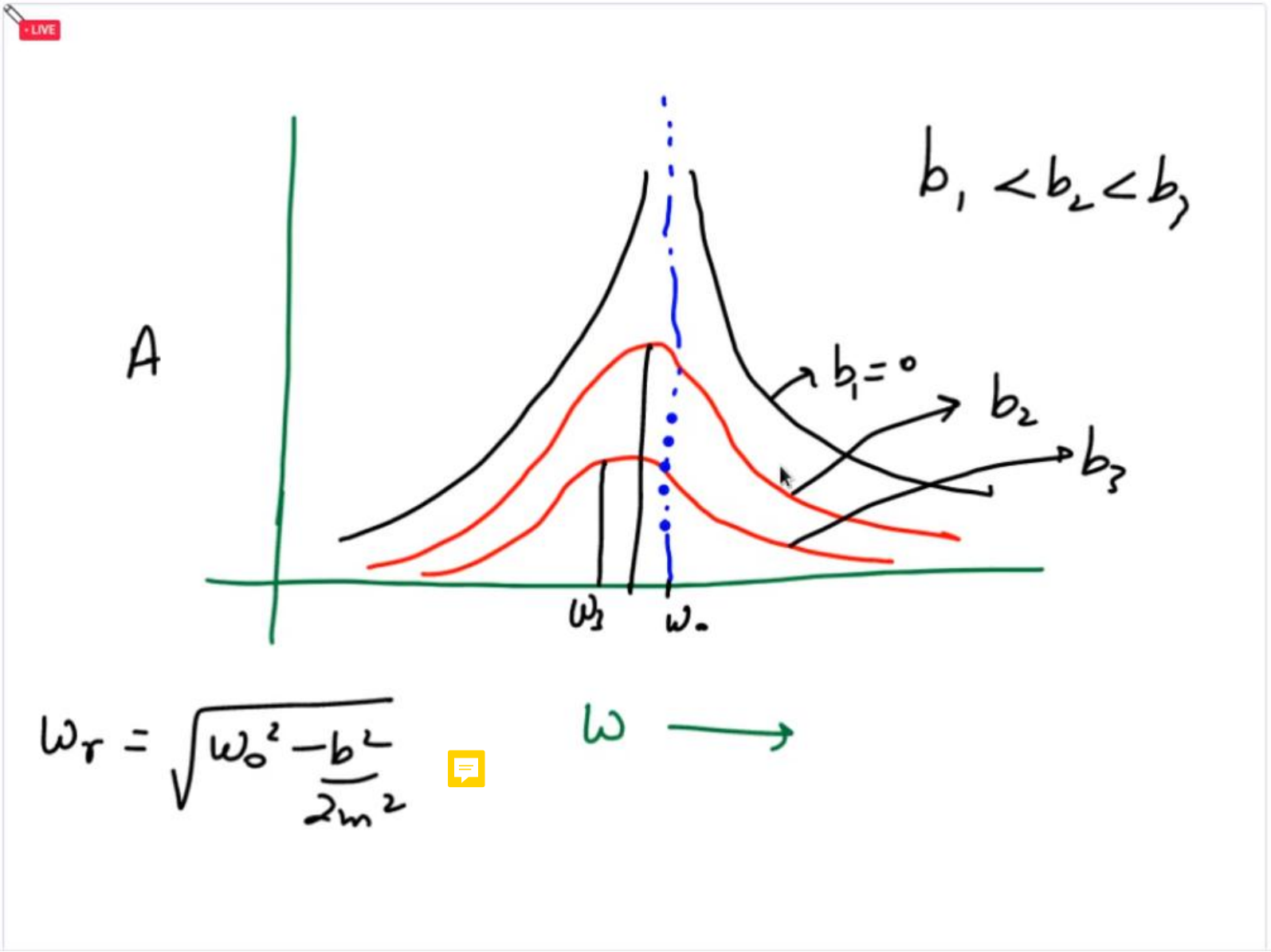
SA Samarth Nagpal
Y

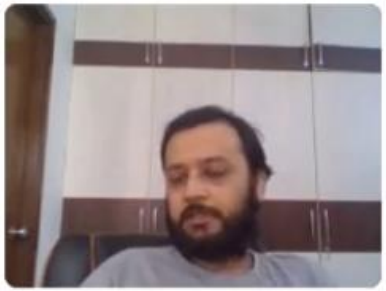
Samarth Nagpal

SA Sanskar Saxena
power max

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Chat Doubts More





Abhinash Chauhan
y

SA Samarth Nagpal
Yes sir

PR Pranesh K.r
yes

PR Pratham omkar Pat...
y

PR Prathik
ok sir

RI Rishav Raj
sir xam me kya karenge

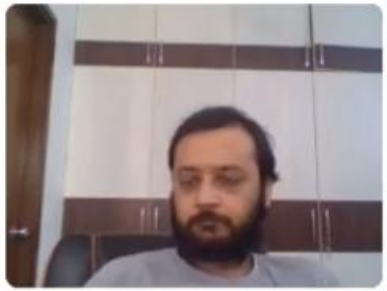
Samarth Nagpal + 1 Other

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Chat Doubts More

LIVE

Q: When a force $F = 10 \sin(2t)$ acts on a particle of $m = 1$ kg it produces a forced oscillation. The velocity amplitude is found to be max. If Angular frequency of force is decreased by 10%. then displacement amplitude becomes max. Find b.



sir xam me kya karenge

Samarth Nagpal + 1 Other

RI Rishav Raj
.

RO ROHIT K
y

PR Pratham omkar Pat...
y

PR Pranesh K.r
y

AB Abhiharsh Chauhan
y

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Chat Doubts More

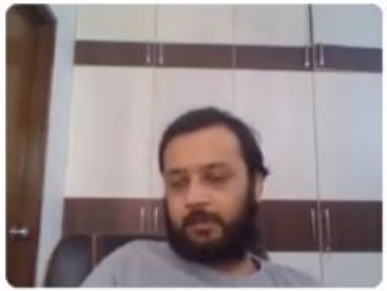
LIVE

Q: When a force $F = 10 \sin(2t)$ (ω) acts on a particle of $m = 1$ kg it produces a forced oscillation. The velocity amplitude is found to be max. If Angular frequency of force is decreased by 10% then displacement amplitude becomes max. Find b.

$\omega_0 = 2$

$\omega_r = 1.8 = \sqrt{\omega_0^2 - \frac{b^2}{2m^2}}$ $\Rightarrow 1.8 = \sqrt{4 - \frac{b^2}{2}}$

$4 - \frac{b^2}{2} =$



Sanyam Kumar Jain
(root19)/5

MA Madhav
0.39

RI Rishav Raj
haa

PR Pranesh K.r
y

PR Prathik
y

AB Abhiharsh Chauhan
y

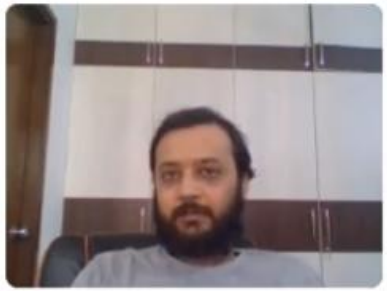
MA Manoj
yes

Type a message

Chat Doubts More

LIVE

Q: A force $F = F_0 \cos(\omega t)$ acts on a damped oscillator to produce forced oscillation. Displacement function is given by $x = A\cos(\omega t - \phi)$. Find the work done by F in one cycle.



Pratheek
y

PR Prathik
y

AB Abhiharsh Chauhan
y

MA Manoj
yes

SA Sanyam kumar Jain
y

HA Harshal Sethiya
0 hai kya??

AR Arnab Goyal
1 min sir

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Chat Doubts More

LIVE

$$x = A\cos(\omega t - \phi)$$
$$v = -A\omega\sin(\omega t - \phi)$$

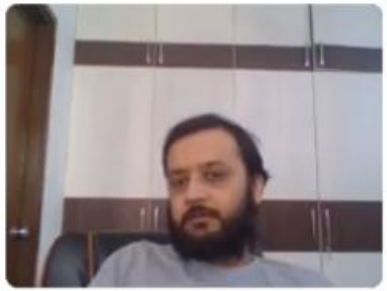
$$dW = P\,dt$$

$$W = \int F\,v\,dt$$

$$= -\int_0^T F_0\cos\omega t\; A\omega\sin(\omega t - \phi)\,dt$$

$$= -F_0 A\omega \int_0^T (\cos\omega t\sin\omega t\cos\phi - \cos^2\omega t\sin\phi)\,dt$$

$$= -F_0 A\omega \left[\cos\phi \int_0^T \frac{1}{2}\sin 2\omega t\,dt - \sin\phi\int_0^T \cos^2\omega t\,dt\right]$$



1 min sir

AB Abhiharsh Chauhan: dhora bada expression aa raha hai

RI Rishav Raj: sir F0(A)pisin(phi)/w

AR Arnab Goyal: 2FAsinphi

AR Arnab Goyal: piFAsinphi

SA Sanskar Saxena: f.ds

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Chat Doubts More

$$x = A\cos(\omega t - \phi)$$
$$v = -A\omega\sin(\omega t - \phi)$$

$$dW = P\,dt$$

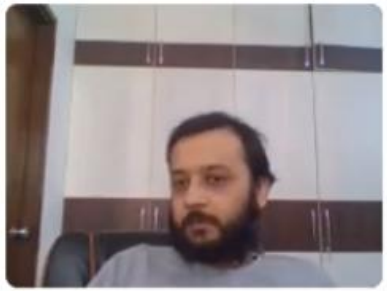
$$W = \int F\,v\,dt$$

$$= -\int_0^T F_0\cos\omega t\; A\omega\sin(\omega t - \phi)\,dt$$

$$= -F_0 A\omega \int_0^T (\cos\omega t\sin\omega t\cos\phi - \cos^2\omega t\sin\phi)\,dt$$

$$= -F_0 A\omega \left[\cos\phi \int_0^T \frac{1}{2}\sin 2\omega t\,dt - \sin\phi \int_0^T \cos^2\omega t\,dt \right]$$

$$\cos^2\omega t = \frac{1+\cos 2\omega t}{2}$$



Arnab Goyal
2FAsinphi

AR Arnab Goyal
piFAsinphi

SA Sanskar Saxena
f.ds

SA Sanskar Saxena
walli formula

Jashan Grover insan

MA Madhav

RI Rishav Raj
cos bhi 0

Type a message

Chat Doubts More

$$x = A\cos(\omega t - \phi)$$

$$v = -A\omega\sin(\omega t - \phi)$$

$$dW = P\,dt$$

$$W = \int F\,v\,dt$$

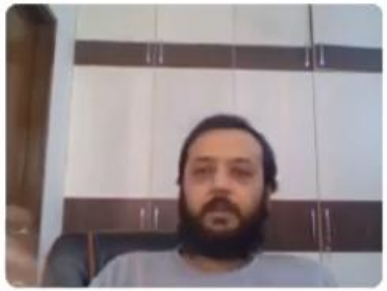
$$= +F_0 A\omega \sin\phi \frac{T}{2} = \boxed{F_0 A\pi \sin\phi}$$

$$= -\int_0^T F_0\cos\omega t\; A\omega \sin(\omega t - \phi)\,dt$$

$$= -F_0 A\omega \int_0^T (\cos\omega t \sin\omega t \cos\phi - \cos^2\omega t \sin\phi)\,dt$$

$$= -F_0 A\omega \left[\cos\phi \int_0^T \frac{1}{2}\sin 2\omega t\,dt - \sin\phi \int_0^T \cos^2\omega t\,dt \right]$$

$$\cos^2\omega t = \frac{1+\cos 2\omega t}{2}$$



Arnab Goyal
piFAsinphi

SA Sanskar Saxena
f.ds

SA Sanskar Saxena
walli formula

Jashan Grover insan

MA Madhav

RI Rishav Raj
cos bhi 0

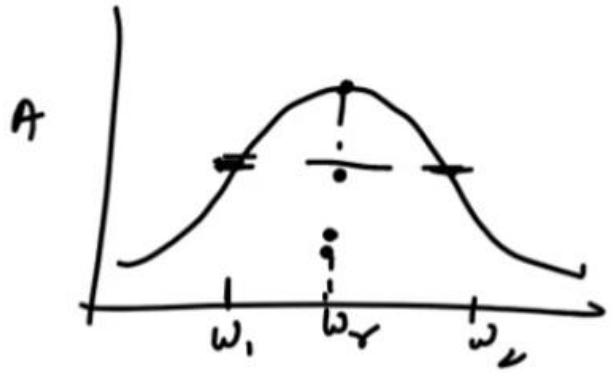
RI Rishav Raj
.

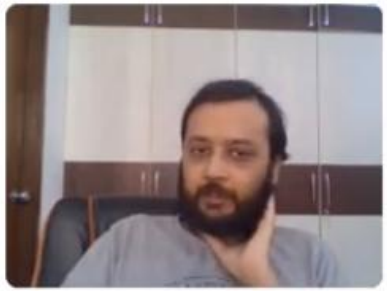
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Chat Doubts More

LIVE

Q: Displacement amplitude in a forced oscillation was found to be equal at $\omega_1 = 400$ and $\omega_2 = 600$. Find the ω at which displacement amplitude will be max.





yees sir

PR Prathik: yes sir

MO Mohit Singh: Yes

AB Abhiharsh Chauhan: I by mistake integrated from 0 to 2pi instead of 0to 2pi/w

PR Pratham omkar Pat...: yes sir

AR Arnab Goyal: 200root6?

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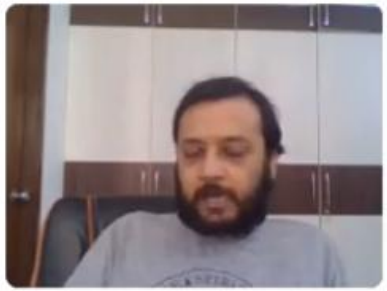
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LIVE

Q: Displacement amplitude in a forced oscillation was found to be equal at $\omega_1 = 400$ and $\omega_2 = 600$. Find the ω at which displacement amplitude will be max.

$$A = \frac{F_0}{\sqrt{b^2\omega^2 + m^2(\omega^2 - \omega_0^2)^2}}$$

$$\omega_r = \sqrt{\omega_0^2 - \frac{b^2}{2m^2}}$$



Arnab Goyal: 200root6?

Madhav: .

Manoj: yes

Arnab Goyal: 100root26

Rishav Raj: sir 100root26

Arnab Goyal: pehle 100 root24 kar diya tha galti se

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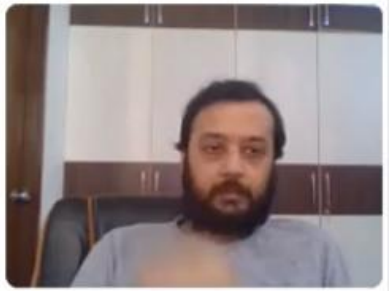
$$b^2 \times 400^2 + m^2(400^2 - \omega_0^2)^2 = b^2 \times 600^2 + m^2(600^2 - \omega_0^2)^2$$

$$b^2(600^2 - 400^2) = -m^2\left[(400^2 - \omega_0^2 - 600^2 + \omega_0^2)\right.$$
$$\left.(400^2 - \omega_0^2 + 600^2 - \omega_0^2)\right]$$

$$b^2 = +m^2\left[-400^2 - 600^2 + 2\omega_0^2\right)$$

$$\frac{b^2}{2m^2} = \omega_0^2 - \frac{(400^2 + 600^2)}{2}$$

$$\sqrt{\omega_0^2 - \frac{b^2}{2m^2}} = \sqrt{\frac{400^2 + 600^2}{2}} = 100\sqrt{26}$$



pehle 100 root24 kar diya tha galti se

JA Jashan Grover Insan
100rt26

AB Abhiharsh Chauhan
100rt26

SA Samarth Nagpal
Put 100 square as k

SA Samarth Nagpal
To reduce cal

MA Madhav
100rt26

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